

our cameras and stopped all pure-play services. This put pressure on our sales, but we managed to get it done.

How many IPs do you own?

There are a lot of IPs that we have worked on, both on the camera side as well as image sensing, image processing, camera interfacing, and mechanical design. Recently, in the last few years, we have started filing patents. We have three US patents which have been granted, and there are eight more which have been filed. There are more in the pipeline.

How do you do your market research when it comes to IPs?

We look at a lot of industry studies and take information from there. We have a very vibrant customer pool built over the years. If there are 300 products in the world that use our cameras, that means we'd have talked to at least 3,000-4,000 customers before we reached that 300. From a camera perspective, we learn what is the next thing that is expected.

We are also very active in the communities. We have many technology partners, such as NVIDIA, and our discussions with our partner ecosystem help us. We are part of the Embedded Vision Alliance. We take part in a lot of embedded vision shows and embedded systems conferences. These are where we get to do our market research and collect information to decide on the next thing we need to do.

How much of your revenue do you dedicate towards research?

I cannot give it as a percentage of our hardcore revenue. We have a core research and development team and a core product development team. We have our imaging processing lab. We are also building our autonomous mobile robot (AMR) lab just to create an atmosphere for the



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R&D people to work with AMR and understand what kind of cameras or imaging technologies are required.

There is always a continuous investment on R&D. Our major expense in terms of engineering is on the R&D side. Once we come up with a product, it's more of a production game where we can have production partners. We rely on our EMS partners and external manufacturers for production. The major expense that we do is on the engineering side.

What are the latest and most sought after technologies in the embedded vision industry, according to you?

If you ask me, there is always a need for better and low-power imaging. The imaging has to keep on improving. People want better quality images in very low light conditions. There are challenges in lighting. So, people are always looking for sensor

quality improvement.

The second thing is interfacing. Today, there is a serial line technology called the GMSL (Gigabit Multimedia Serial Links). What it means is that I can have the camera 10 or 15 metres away from the processor. This was a big challenge earlier. People had to either take a USB or an Ethernet. This GMSL technology is a serial language that has much higher speed and reliability as well.

But the technology was tough; we worked on it for at least five to six months before we could make a stable product. This technology is required for robots having bigger arms and AMRs. They need a rugged cable. So, there is a requirement for rugged cameras when it comes to imaging quality.

There is also a requirement for cameras that can sense depth. For vision, we have stereo cameras with two imaging sensors, which act like our eyes. I cannot figure